# DN16 Directional Valves

Selection data & performance | Manufacturer technical reference.

### Pilot Operation

Actuation by water, oil or compressed air pilot is possible on all configurations. The water version is also available with an integral DN3 solenoid operated pilot valve. Please note: adequate flow must be available to ensure the full travel of the valve is made. The pilot piston can be sized to vary the pilot actuation pressure between 2 and 80 bar. The external pilot actuator is designed to operate at a pressure of 40 bar, although alternatives can be offered on request. For oil pilot, the actuator is supplied with a dual seal and air vent design to prevent cross contamination of the fluids.

### Return Options

The valves shown are all spring return however, the valves can be offered with a detent function and pilot return if preferred.

| SPECIFICATION | |
|---|---|
| Max Inlet Pressure | 160 bar |
| Operating Pressure Range | 0 to 100 bar |
| Flow Range | 0-140 L/min |
| Flow Characteristics | CV Rate 5.0<br>Kv Rate 4.26 |
| Feed Gallery Diameter | 18mm |
| Porting | 3/4" BSP (Parallel) |
| Construction Materials | A.I.S.I 316, Polymer & Martensitic Stainless steel |
| Pilot Operation | See Note |
| Pilot Media | Water, Air & Oil |

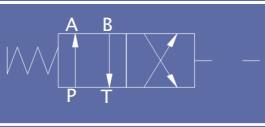
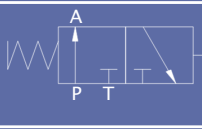
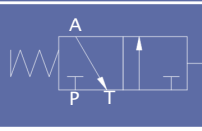


**Functions:**
4/2
3/2 N/O
3/2 N/C

The size of the pilot actuation can be modified to generate an actuation of different pilot pressure. For specific operating range contact TWHC. The DN16 is also available with an integral 3/2 solenoid operated pilot actuation valve. See 20*EESW for basic dimensions.

### Electrical Actuation

Due to actuation forces and stroke length a direct solenoid operation is not possible with this size of valve. However, stepper motor control is under development.

| ORDERING CODES | Ext. Water Pilot | Oil Pilot | Air Pilot |
|---|---|---|---|
|  | 201EJSW | 201ELSW | 201EFSW |
|  | 208EJSW | 208ELSW | 208EFSW |
|  | 203EJSW | 203ELSW | 203EFSW |

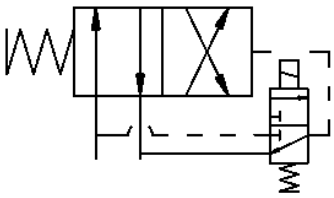
Size from actual flow and pressure drop. Confirm pilot supply, tank return, spool function, electrical supply and the selected manifold interface. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

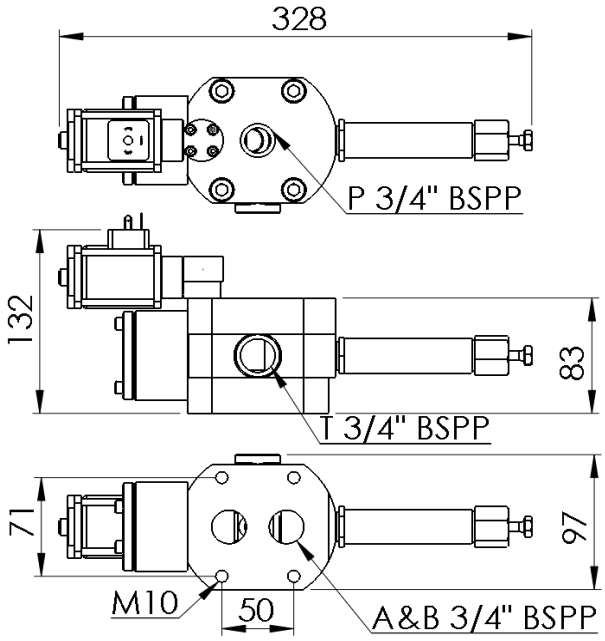
DN16 Directional Valves V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

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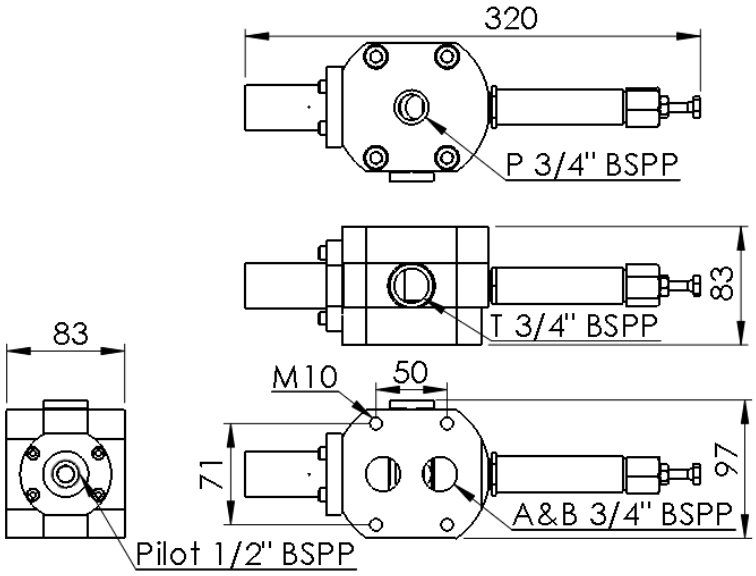
Dimensions, connections & configuration | Manufacturer technical reference.

### 201EDSW & 201EESW






### 201EJSW , 201ELSW & 201EFSW




Actuation housing size may vary with type of pilot and operating pressures.4/2 valve shown but also available in 3/2 with N/O or N/C

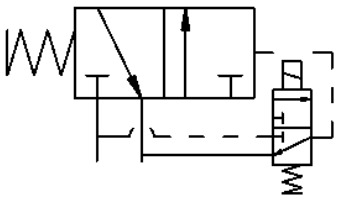
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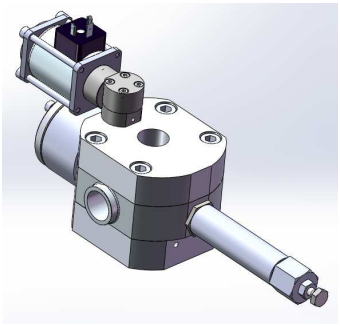
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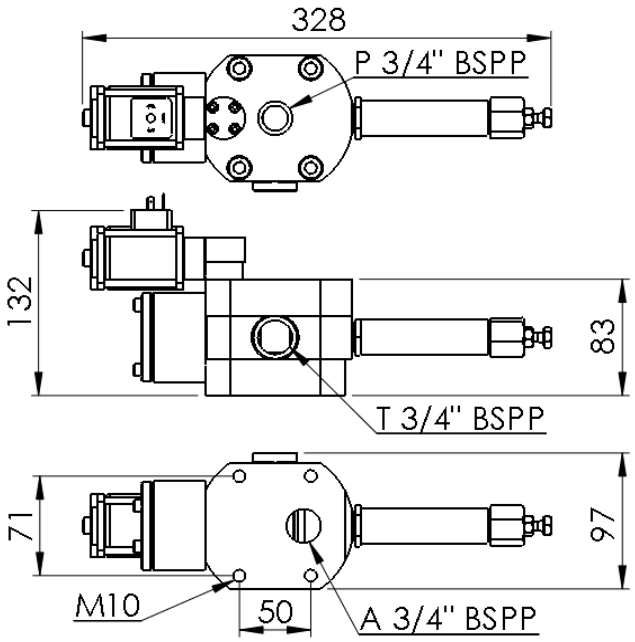
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Dimensions, connections & configuration | Manufacturer technical reference.

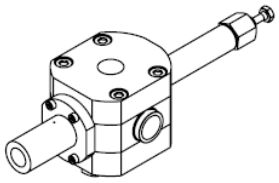
### 203EDSW & 203EESW

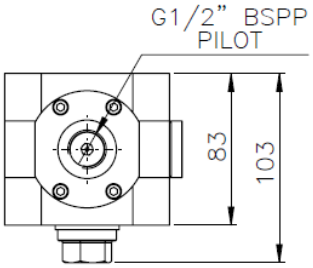


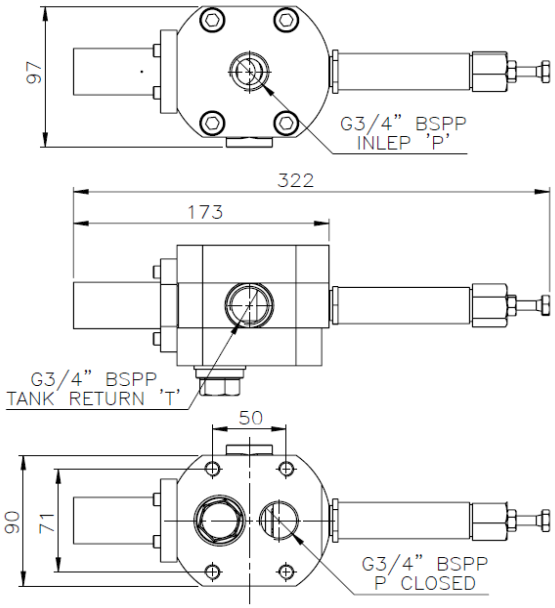





### 203EJSW , 203ELSW & 203EFSW





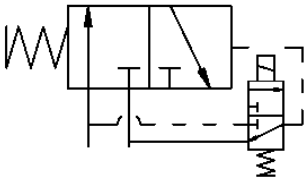

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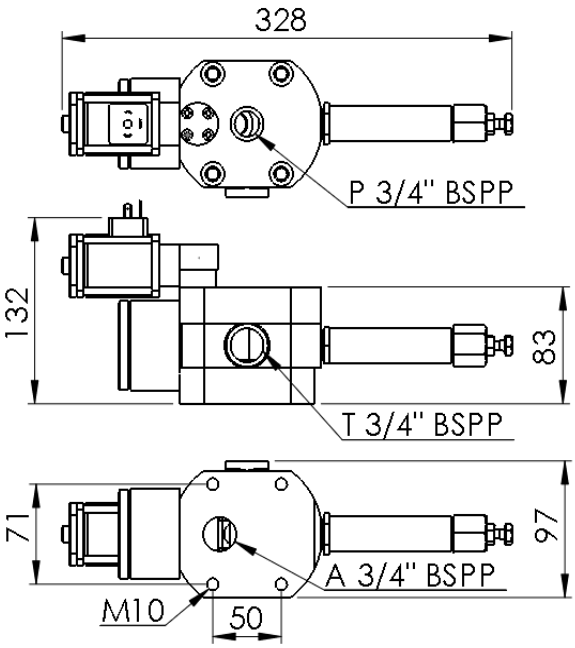
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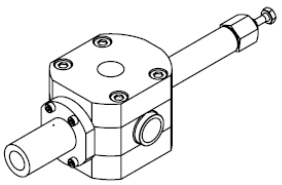
Dimensions, connections & configuration | Manufacturer technical reference.

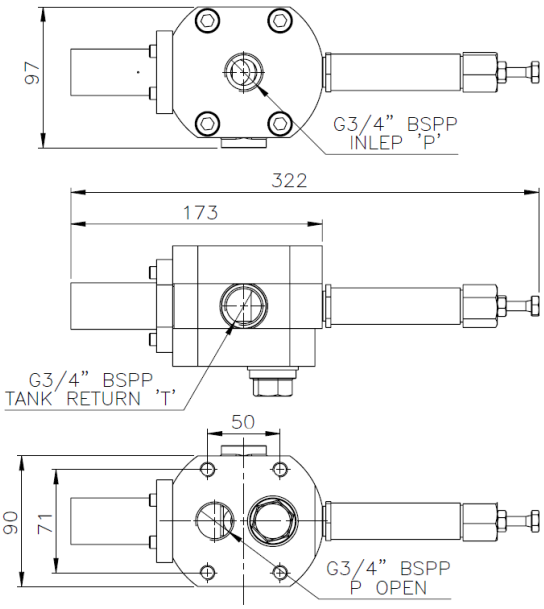
### 208EDSW & 208EESW

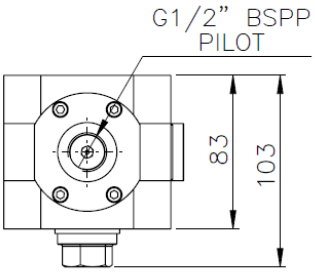





### 208EJSW , 208ELSW & 208EFSW






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